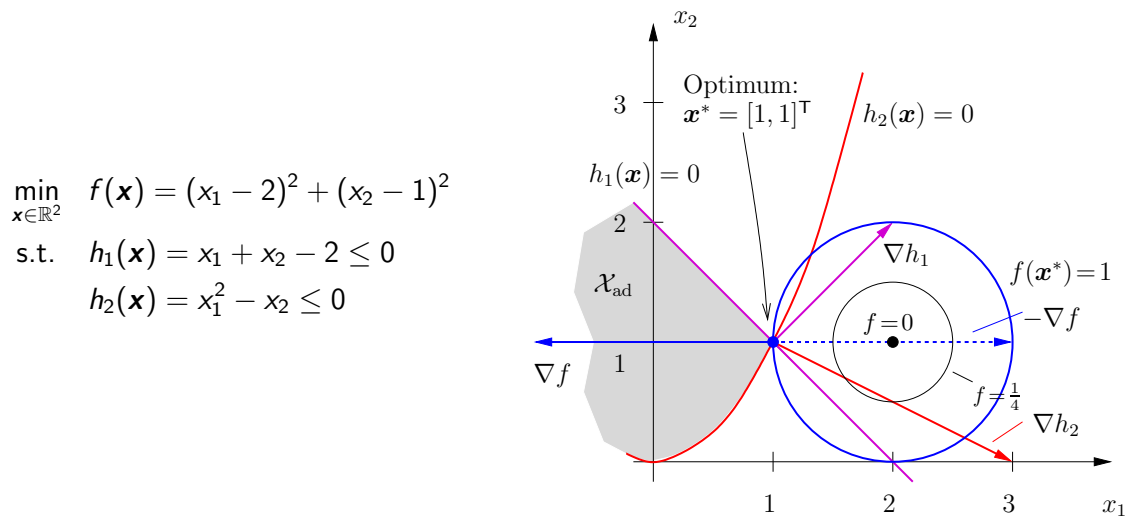


Solution to Exercise 4.1 (Page 48)



- Admissible set $\mathcal{X}_{ad} = \{\mathbf{x} \in \mathbb{R}^2 : h_1(\mathbf{x}) \leq 0, h_2(\mathbf{x}) \leq 0\}$

- Optimal solution $\mathbf{x}^* = [1, 1]^T$ lies on the edge of $\mathcal{X}_{ad}$

$\Rightarrow$ Both constraints are active: $h_1(\mathbf{x}^*) = 0, h_2(\mathbf{x}^*) = 0$ [see figure]

$\Rightarrow$ Complementary cond. $\mu_1^* h_1(\mathbf{x}^*) = 0$ and $\mu_2^* h_2(\mathbf{x}^*) = 0$ in (4.22) are satisfied

- Gradients $\nabla f(\mathbf{x}) = \begin{bmatrix} 2(x_1 - 2) \\ 2(x_2 - 1) \end{bmatrix}$, $\nabla h_1(\mathbf{x}) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\nabla h_2(\mathbf{x}) = \begin{bmatrix} 2x_1 \\ -1 \end{bmatrix}$
- Lagrange condition $\nabla_x L(\mathbf{x}^*, \boldsymbol{\mu}^*) = 0$ in (4.22)

$$\nabla_x L(\mathbf{x}^*, \boldsymbol{\mu}^*) = \begin{bmatrix} -2 \\ 0 \end{bmatrix} + \mu_1^* \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \mu_2^* \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \mathbf{0}$$

yields $\mu_1^* = \frac{2}{3}, \mu_2^* = \frac{2}{3}$

$\Rightarrow$ Necessary opt. cond. (4.22) are satisfied for $\mathbf{x}^*$

- Illustration: $-\nabla f(\mathbf{x}^*)$ lies in the cone defined by $\nabla h_1(\mathbf{x}^*), \nabla h_2(\mathbf{x}^*)$

- In Section 4.2.1: $\nabla \mathbf{h}_{\mathcal{A}(\mathbf{x}^*)}(\mathbf{x}^*) = \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix}$ with $|\cdot| = -1 - 2 = -3$

$\Rightarrow$ Constraint qualification is satisfied

Solution to Exercise 4.2 (Page 51)

- Lagrangian function

$$L(\mathbf{x}, \boldsymbol{\mu}) = \overbrace{(x_1 - 2)^2 + \frac{1}{9}(x_2 - 2)^2}^{= f(\mathbf{x})} + \mu_1 \overbrace{(2x_1 + x_2 - 2)}^{= h_1(\mathbf{x})} + \mu_2 \overbrace{(x_1 - x_2 - 1)}^{= h_2(\mathbf{x})}$$

$$\nabla_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\mu}) = \begin{bmatrix} 2(x_1 - 2) \\ \frac{2}{9}(x_2 - 2) \end{bmatrix} + \mu_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \mu_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Cases	Constraints	CC (4.29e)	Eqns.	Solution	Stat. point?
a) unconstr.	$h_1(\mathbf{x}) < 0$ $h_2(\mathbf{x}) < 0$	$\mu_1 = 0$ $\mu_2 = 0$	$\nabla_{\mathbf{x}} L(\mathbf{x}) = \mathbf{0}$	$\mathbf{x} = [2, 2]^T$	no ($\mathbf{x} \notin \mathcal{X}_{ad}$)
b) h_1 active	$h_1(\mathbf{x}) = 0$ $h_2(\mathbf{x}) < 0$	$\mu_1 = ?$ $\mu_2 = 0$	$\begin{bmatrix} \nabla_{\mathbf{x}} L(\mathbf{x}, \mu_1) \\ h_1(\mathbf{x}) \end{bmatrix} = \mathbf{0}$	$\mathbf{x} = \left[\frac{18}{13}, -\frac{10}{13}\right]^T$ $\mu_1 = \frac{8}{13}$	no ($\mathbf{x} \notin \mathcal{X}_{ad}$)
c) h_2 active	$h_1(\mathbf{x}) < 0$ $h_2(\mathbf{x}) = 0$	$\mu_1 = 0$ $\mu_2 = ?$	$\begin{bmatrix} \nabla_{\mathbf{x}} L(\mathbf{x}, \mu_2) \\ h_2(\mathbf{x}) \end{bmatrix} = \mathbf{0}$	$\mathbf{x} = \left[\frac{21}{10}, \frac{11}{10}\right]^T$ $\mu_2 = -\frac{1}{5}$	no ($\mathbf{x} \notin \mathcal{X}_{ad}, \mu_2 < 0$)
d) h_1, h_2 active	$h_1(\mathbf{x}) = 0$ $h_2(\mathbf{x}) = 0$	$\mu_1 = ?$ $\mu_2 = ?$	$\begin{bmatrix} \nabla_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\mu}) \\ h_1(\mathbf{x}) \\ h_2(\mathbf{x}) \end{bmatrix} = \mathbf{0}$	$\mathbf{x} = [1, 0]^T$ $\boldsymbol{\mu} = \left[\frac{22}{27}, \frac{10}{27}\right]^T$	yes: $\mathbf{x}^* = \mathbf{x}$

- Example case b):

$$\begin{cases} 2(x_1 - 2) + 2\mu_1 = 0 & (1) \\ \frac{2}{9}(x_2 - 2) + \mu_1 = 0 & (2) \\ 2x_1 + x_2 - 2 = 0 & (3) \end{cases}$$

from (3): $x_2 = 2 - 2x_1$

with (2): $\mu_1 = -\frac{2}{9}(2 - 2x_1 - 2) = \frac{4}{9}x_1$

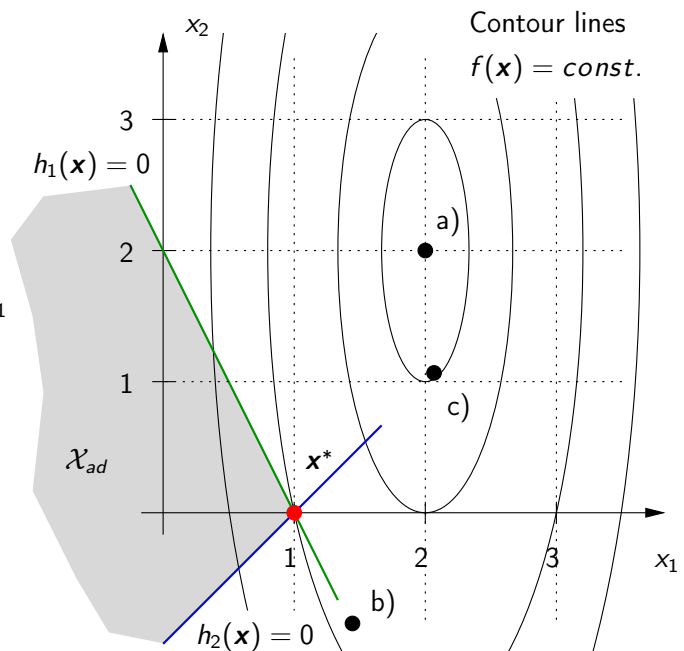
with (1): $2(x_1 - 2) + \frac{8}{9}x_1 = 0$

$$\Rightarrow x_1 = \frac{18}{13}$$

$$x_2 = 2 - \frac{36}{13} = -\frac{10}{13}$$

$$\mu_1 = \frac{4}{9} \cdot \frac{18}{13} = \frac{8}{13}$$

- Illustration:



Solution to Exercise 4.3 (Page 51)

$$\bullet L(\mathbf{x}, \boldsymbol{\mu}) = (x_1 - \frac{3}{2})^2 + (x_2 - s)^4 + \underbrace{\mu_1(-1 + x_1 + x_2)}_{h_1(\mathbf{x})} + \underbrace{\mu_2(-1 + x_1 - x_2)}_{h_2(\mathbf{x})} + \underbrace{\mu_3(-1 - x_1 + x_2)}_{h_3(\mathbf{x})} + \underbrace{\mu_4(-1 - x_1 - x_2)}_{h_4(\mathbf{x})}$$

$$\bullet \nabla_{\mathbf{x}} L(\mathbf{x}^*, \boldsymbol{\mu}) = \begin{bmatrix} 2(x_1 - \frac{3}{2}) + \mu_1 + \mu_2 - \mu_3 - \mu_4 \\ 4(x_2 - s)^3 + \mu_1 - \mu_2 + \mu_3 - \mu_4 \end{bmatrix} = \mathbf{0}$$

$$\bullet \text{ At point } \mathbf{x} = [1, 0]^T \text{ we have } h_1(\mathbf{x}^*) = 0, h_2(\mathbf{x}^*) = 0, h_3(\mathbf{x}^*) < 0, h_4(\mathbf{x}^*) < 0$$

$$\Rightarrow \mu_3 = \mu_4 = 0$$

$$\bullet 2(x_1 - \frac{3}{2}) + \mu_1 + \mu_2 = 0 \text{ and } 4(x_2 - s)^3 + \mu_1 - \mu_2 = 0 \text{ give}$$

$$\mu_1 = \frac{1 + 4s^3}{2}, \quad \mu_2 = \frac{1 - 4s^3}{2}$$

$$\bullet \text{ The KKT sign condition } \mu_1, \mu_2 \geq 0, \text{ see (4.29d), is satisfied for}$$

$$-\sqrt[3]{1/4} \leq s \leq \sqrt[3]{1/4}$$

Solution to Exercise 4.4 (Page 52)

$$\bullet \text{ Theorem 4.3: Hessian } \nabla_{\mathbf{x}\mathbf{x}}^2 L(\mathbf{x}^*, \boldsymbol{\mu}^*) = \begin{bmatrix} 2 & 0 \\ 0 & \frac{2}{9} \end{bmatrix} \text{ is positive definite}$$

$$\Rightarrow \mathbf{s}^T \nabla_{\mathbf{x}\mathbf{x}}^2 L(\mathbf{x}^*, \boldsymbol{\mu}^*) \mathbf{s} > 0 \text{ is satisfied } \forall \mathbf{s} \neq \mathbf{0}$$

$$\Rightarrow \mathbf{x}^* \text{ is strict local minimum point}$$

$$\bullet \text{ Theorem 4.4: } f(\mathbf{x}) \text{ is (strictly) convex, } h_1(\mathbf{x}) \text{ \& } h_2(\mathbf{x}) \text{ are linear (and thus convex)}$$

$$\Rightarrow \mathbf{x}^* \text{ is a global minimum point}$$

Solution to Exercise 4.5 (Page 53)

$$\bullet \mu_1^* = 22/27 \approx 0.815 \text{ and } \mu_2^* = 10/27 \approx 0.370 \text{ for } \mathbf{x}^* = [1, 0]^T$$

$$\Rightarrow \text{Change/relaxation of } h_1(\mathbf{x}) \leq \varepsilon_1 \text{ has stronger effect on } f \text{ than } h_2(\mathbf{x}) \leq \varepsilon_2 \text{ (see Figure 4.6).}$$